

SMART INDIA HACKATHON 2026



- **Problem Statement ID - 26090**
- **Problem Statement Title - AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans**
- **Theme - Heritage and Culture**
- **PS Category - Software**
- **Team ID -**
- **Team Name - Hacktastic Six**



KalaSetu is an AI-powered operations assistant that enables artisans to photograph, price, list, and fulfill orders entirely through voice, without literacy, technical skill, or reliable connectivity.

How It Addresses The Problem

NO PHOTO SKILLS

AI Image Studio

Turns any phone photo into a clean, marketplace-ready product image automatically.

LANGUAGE BARRIER

Voice-to-Listing

Speak in your language and get a bilingual, SEO-ready listing in seconds.

PRICING EXPLOITATION

Fair-Price Engine

Enforces a cost floor and benchmarks against live market rates so there is no more selling below cost.

NO CONNECTIVITY

Offline Capture

Work continues with no signal; everything syncs and publishes once you're back online.

LACK OF TRUST & LITERACY

Voice Confirmation

Every listing, price, and order detail is read back aloud. Artisans approve entirely by ear, no reading or typing required.



A multistage CV pipeline and craft-glossary-biased transcription work together to produce studio-ready images and SEO-optimized bilingual listings. Fully offline-capturable, synced on reconnect.

Innovation & Uniqueness

Full Order Lifecycle

Beyond generating listings, the platform supports post-sale work too. Artisans can create social media posts, manage incoming orders, get packaging guidance by category, and generate a personal-story label for shipments, turning each sale into a branded experience

Craft-Aware Transcription

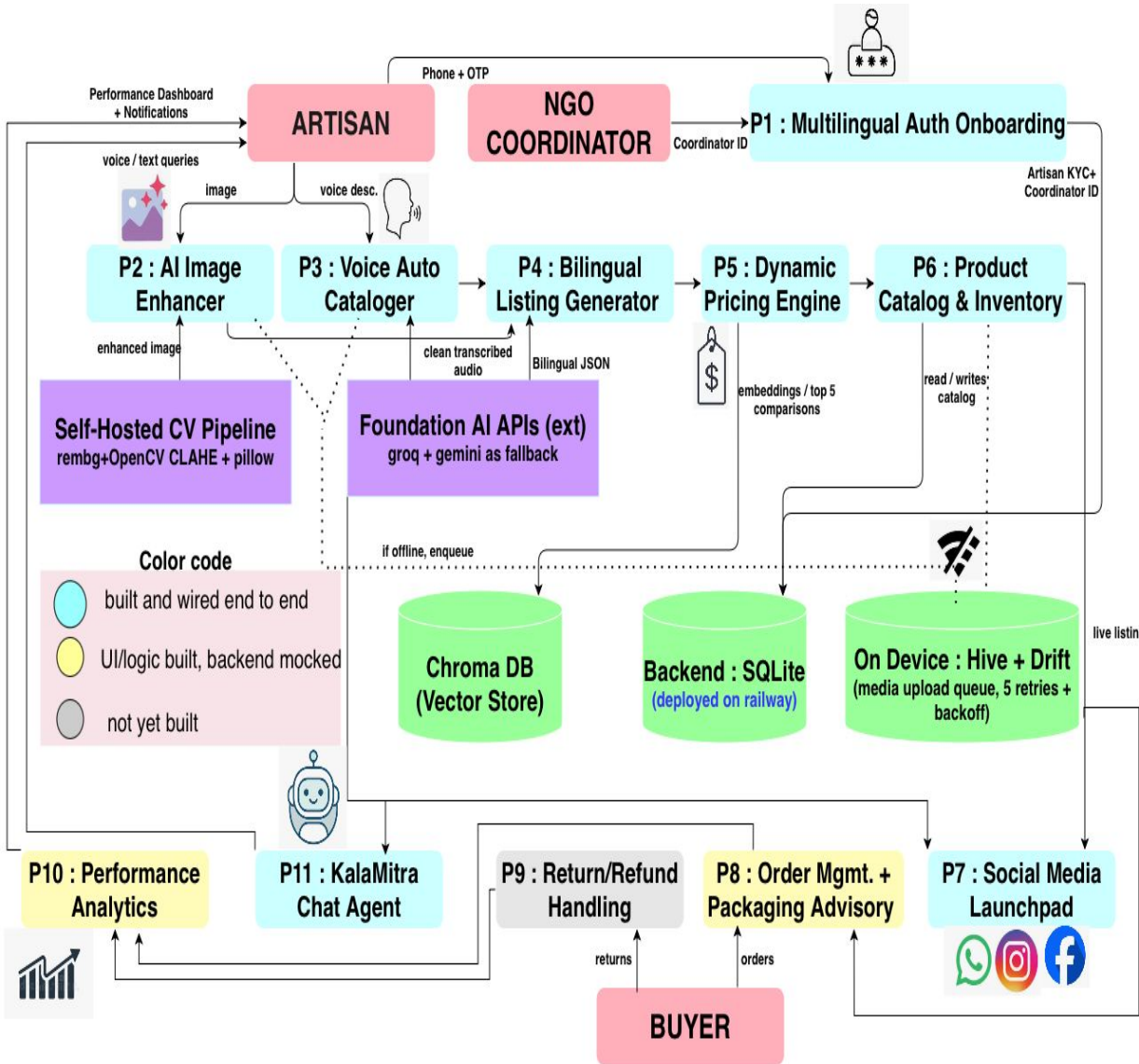
Voice recognition tuned with a craft-glossary bias, so regional terms for materials and techniques (e.g. "bandhani," "dhokra") are captured accurately, a gap unaddressed by general-purpose speech-to-text systems.

No Seller-Side Commission

Core selling tools are free permanently; monetization happens only on the B2B/institutional side, so unit economics don't depend on taxing the artisan.

Conversational Business Interface

Rather than navigating separate screens, artisans can interact with their business through a single multilingual voice/text assistant - KalaMitra, removing the need to read or operate a traditional app interface.



Architectural Decisions

Flutter 3.x + Riverpod 2.x: Native performance on low-cost Android devices with direct hardware access for camera and microphone.

Drift (SQLite) + Hive + WorkManager: OS-level background sync drains queued uploads even after the app is closed, ensuring no work is lost to connectivity gaps.

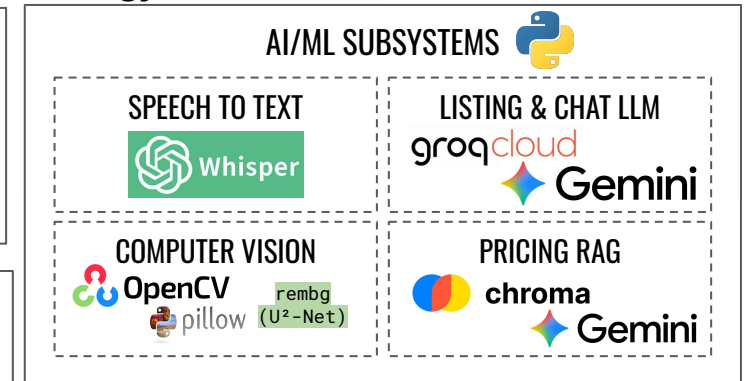
Rembg (U²-Net) + OpenCV (CLAHE) + Pillow: Self-hosted image processing avoids per-call cloud vision costs and corrects poor rural lighting conditions.

Groq Cloud (Whisper v3) + Gemini Fallback: Sub-second inference is critical for a voice-first UX; dual-provider routing ensures resilience if one service fails.

Deterministic Cost Formula + ChromaDB RAG: Prevents LLM price hallucination by anchoring every AI suggestion to a hard-coded cost floor, cross-checked against indexed comparables scraped from Amazon Karigar, Etsy, and FabIndia.

FastAPI + Uvicorn + Railway PaaS: Asynchronous REST API with zero-downtime continuous deployment.

Technology Stack & Platforms



IMPACTS AND BENEFITS



Year-Round Revenue

Converts seasonal mela-cycle income into consistent revenue across the calendar year, supported by an enforced price floor that prevents underselling.



Female Empowerment

Enables home-based business operation, removing location and mobility as barriers to participation.



Verifiable Sales History

Each transaction generates a digital record of income — a data foundation that cash-based artisans currently lack, and a prerequisite for future formal credit access.



No Middlemen

Removes dependence on intermediaries who capture 3–5x sale value, enabling direct payouts and, where credit-linked relationships previously existed, greater financial independence.



Indigenous Heritage Preservation

Voice-based cataloging captures technique and provenance details at the point of sale, creating a searchable digital record of craft knowledge that would otherwise remain undocumented.



Expanded Market Access

Extends an artisan's customer base beyond local and seasonal markets through direct integration with WhatsApp, Instagram, and Facebook.



Institutional Data Visibility

Aggregated, anonymized sales and inventory data from craft clusters can inform state handicraft board planning, the basis for institutional SaaS licensing as a sustainability channel.

Future Scope

Native ONDC & WhatsApp: Live onboarding to the ONDC network and GeM portal for zero commission sales, paired with a WhatsApp bot for easy photo/voice cataloging.

Bhashini Dialect Expansion: Upgrading voice AI to recognize tribal and indigenous dialects (like Santhali and Gondi), bringing remote craftspeople online.

Pehchan ID & Credit: Enabling one tap DigiLocker onboarding via Pehchan Card to instantly unlock PM Vishwakarma credit (up to ₹3 Lakh) with alternative scoring.

Global Export Automation: Auto mapping listings to HS Codes and generating postal export declarations (CN22/CN23) for seamless India Post shipping.

Multi-Artisan SHG Hubs: Empowering coordinators to manage shared digital inventory for Self-Help Groups, automatically routing payouts to individual accounts.

Feasibility

- **Low Technical Risk:** Built on proven components (rembg, Whisper) for core CV/STT tasks, with hosted LLM APIs (Groq/Gemini) for language generation. No novel model training required.
- **Institutional Alignment:** Designed for direct integration with ONDC and GeM; Bhashini-based dialect support planned as a subsequent phase.
- **Modular Architecture:** All independently developed processes enable incremental delivery and reduce the risk of a single point of failure across the system.

CHALLENGES

Regional language diversity may reduce transcription accuracy.

Poor/low internet access in rural artisan clusters.

Low digital trust among artisans.

Low-literacy artisans cannot verify AI descriptions/pricing.

B2B buyer trust in new, unverified rural sellers.

Reliance on third-party APIs (Groq/Gemini) adds cost/latency.

MITIGATION STRATEGIES

→ Glossary biasing + 10 Indian languages supported + re-record fallback.

→ Offline-first capture with automated local queue sync on reconnect.

→ NGO & cluster-coordinator assisted in-app onboarding.

→ TTS voice readback + mandatory coordinator review before live.

→ NGO sign-off, with GI-tag/craft-certification badges displayed where applicable.

→ Self-hosted OSS models as primary fallback; Groq/Gemini dual-provider routing for load balancing and cost optimization.

Sustainability Strategy

Revenue is deliberately NOT extracted from artisans.

Free Core Tier: Cataloging, pricing, and listing publication are free for every artisan, permanently. Subsidized by the revenue streams below, never billed to the artisan.

B2B commission: 2–5% on bulk/institutional transactions, charged on the buyer side of the marketplace.

Government SaaS: Licensing to state handicraft boards that already fund artisan training and exhibitions.

CSR & grants: Heritage-preservation and livelihood funding underwrites onboarding kiosks and coordinator time.

Core Model Lineage

- Qin et al., U²-Net (2020) architecture behind rembg's background removal.
- Groq's Whisper Large v3 docs
- danielgatis/rembg, xuebinqin/U-2-Net GitHub repository reference implementations

Government/DPI alignment

- PIB press release on ONDC as Digital Public Infrastructure
- NITI Aayog, "DPI@247" — India's own current strategic roadmap for using DPI to drive inclusive growth, the government direction KalaSetu is aligning with.

Pricing Methodology

Fairwork (Oxford Internet Institute) — rates platforms on Fair Pay, among other principles, specifically to stop algorithms pricing workers below a living minimum.

Competitive landscape

Amazon Karigar, Flipkart Samarth, GoCoop, ListIQ

UX/voice-interface justification

- Patel, Rajput et al., "Experiences Designing a Voice Interface for Rural India" (Avaaj Otalo) — supports pairing voice input with confirmation UI rather than voice-only
- Designing Mobile interfaces for Novice and Low-Literacy Users — Medhi, Patnaik, Brunskill, Gautama, Thies, Toyama, ACM ToCHI 2011

